

EGFR fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes EGFR gene fusions from the msk_impact_50k_2026 study, covering 55 fusion events. 28/55 fusions (50.9%) were in-frame. 41/55 fusions (74.5%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test ($p=0.0114539$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for EGFR.

1 Results summary

1.1 Fusion partner frequency

Fusion-partner frequency was tabulated across 55 analyzed fusion events, identifying 31 distinct fusion partner genes. The most frequent partner was SEPTIN14, observed in 18 events.

1.2 Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 25/28 (89.3%) of in-frame fusions retained the domain, $p=0.0114539$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.247525.

1.3 Domain disruption

Domain-disruption analysis was skipped: EGFR has no disruption_required_domains configured; domain-disruption analysis was skipped.

1.4 Cutpoint detection

Cutpoint detection scanned 55 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 982 aa (permutation-corrected $p=0.128713$, not statistically significant at $\alpha=0.05$). This is 17 aa from the nearest configured domain boundary at 965 aa.

1.5 Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: EGFR has no group_field configured; confidence-stats analysis was skipped.

1.6 Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

1.7 Exon retention

Exon retention reported: No target exon was supplied in params or GeneConfig.

1.8 Joint partner

Joint partner reported: EGFR has no gene_pair configured; joint-partner analysis was skipped.

2 Tables

2.1 composite_score composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
SEPTIN14	18	0.32727272727272727	0.060638136511060495		0.9786998132586368		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.3716376061405412	1
RAD51	2	0.03636363636363636	0.060638136511060495		0.9989495798319328		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2513015171594942	2
LANCL2	3	0.05454545454545454	0.060638136511060495		0.9618347338935574		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.25114054367919303	3
ELAPOR2	1	0.01818181818181818	0.060638136511060495		0.9989495798319328		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2435093093672864	4
PKD1L1	1	0.01818181818181818	0.060638136511060495		0.9989495798319328		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2435093093672864	5
VPS41	1	0.01818181818181818	0.060638136511060495		0.9957983193277311		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2428340392592432	6
ZNF713	1	0.01818181818181818	0.060638136511060495		0.9401260504201681		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2309042673504797	7
VWC2	2	0.03636363636363636	0.060638136511060495		0.9028361344537815		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2307057788641761	8
NIPSNAP2	3	0.05454545454545454	0.060638136511060495		0.8658963585434174		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2305823203898773	9
TNRC18	1	0.01818181818181818	0.060638136511060495		0.9233193277310925		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2273028267742492	10
BLTP3B	1	0.01818181818181818	0.060638136511060495		0.9023109243697479		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.22280102605396107	11
CDH7	1	0.01818181818181818	0.060638136511060495		0.832983193277311		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2079450836770103	12
FADD	2	0.03636363636363636	0.060638136511060495		0.6901260504201681		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.18512504657125892	13
EEA1	1	0.01818181818181818	0.060638136511060495		0.6911764705882353		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.17755792881506555	14
KIF5B	1	0.01818181818181818	0.060638136511060495		0.6911764705882353		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.17755792881506555	15
VSTM2A	1	0.01818181818181818	0.060638136511060495		0.6911764705882353		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.17755792881506555	16
CADPS	1	0.01818181818181818	0.060638136511060495		0.6901260504201681		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.17733283877905112	17
NUMA1	1	0.01818181818181818	0.060638136511060495		0.6901260504201681		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.17733283877905112	18
PDE1C	1	0.01818181818181818	0.060638136511060495		0.6523109243697479		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.16922959748253252	19
YIF1B	1	0.01818181818181818	0.060638136511060495		0.6407563025210083		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.16675360708637407	20
TUT7	1	0.01818181818181818	0.060638136511060495		0.6271008403361344		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.1638274366181868	21
CSF2RA	1	0.01818181818181818	0.060638136511060495		0.3214285714285714		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.09832623613799471	22
DDC	1	0.01818181818181818	0.060638136511060495		0.2804621848739496		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.08954772473343288	23
GARS1	1	0.01818181818181818	0.060638136511060495		0.2804621848739496		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.08954772473343288	24

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty	Components_applicable	Composite_score	Rank
SEL1L	1	0.018181818181818	0.060638136511060495		0.2804621848739496		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.08954772473343288	25
TNS3	1	0.018181818181818	0.060638136511060495		0.2804621848739496		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.08954772473343288	26
PCDH15	1	0.018181818181818	0.060638136511060495		0.2710084033613446		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.08752191440930324	27
VOPP1	1	0.018181818181818	0.060638136511060495		0.23004201680672265		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.07874340300474139	28
SCAF4	1	0.018181818181818	0.060638136511060495		0.11764705882352944		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.05465876915119999	29
ZBPB	1	0.018181818181818	0.060638136511060495		0.006302521008403339		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.030799225333672974	30
PLOD3	1	0.018181818181818	0.060638136511060495		0.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.029448685117586543	31

2.2 cutpoint_detection cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
30	1	40	0	14		1.0	-0.0
36	2	39	0	14		1.0	-0.0
142	3	38	0	14		0.5624166190203925	0.24994185469837954
249	3	38	1	13	1.0263157894736843	1.0	-0.0
288	3	38	2	12	0.47368421052631576	0.5924468510484049	0.22735060452753406
297	7	34	2	12	1.2352941176470589	1.0	-0.0
336	8	33	2	12	1.4545454545454546	1.0	-0.0
599	9	32	2	12	1.6875	0.7093311767774617	0.14915095146564436
627	9	32	3	11	1.03125	1.0	-0.0
640	10	31	3	11	1.1827956989247312	1.0	-0.0
651	11	30	3	11	1.3444444444444446	1.0	-0.0
687	13	28	5	9	0.8357142857142857	1.0	-0.0
688	15	26	6	8	0.7692307692307693	0.7548366596998944	0.12214701587495107
823	15	26	7	7	0.5769230769230769	0.528760430668828	0.27674105235972185
875	15	26	9	5	0.32051282051282054	0.1175979559898796	0.9296002268145651
981	15	26	10	4	0.23076923076923078	0.031971018585759146	1.4952435270336717
982	21	20	12	2	0.175	0.02895546314423952	1.5382694841107174
983	31	10	13	1	0.23846153846153847	0.25480492869908067	0.5937921756847577
986	32	9	13	1	0.27350427350427353	0.4229368950729017	0.37372442734387856
994	33	8	13	1	0.3173076923076923	0.4208384837822578	0.3758845527657329
1003	34	7	13	1	0.37362637362637363	0.663943317750199	0.17786899569082928
1039	35	6	13	1	0.44871794871794873	0.663943317750199	0.17786899569082928
1054	37	4	13	1	0.7115384615384616	1.0	-0.0
1055	38	3	14	0	0.0	0.5624166190203926	0.24994185469837946
1060	39	2	14	0	0.0	1.0	-0.0
1075	40	1	14	0	0.0	1.0	-0.0

2.3 domain_retention frame_domain_contingency_table

25	16
3	11

2.4 domain_retention domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained
kinase	Protein tyrosine and serine/threonine kinase	55	41	3	11	0.10304912478825523

2.5 frequency Partner_gene_counts

Partner_gene	Event_count
BLTP3B	1
CADPS	1
CDH7	1
CSF2RA	1
DDC	1
EEA1	1
ELAPOR2	1
FADD	2
GARS1	1
KIF5B	1
LANCL2	3
NIPSNAP2	3
NUMA1	1
PCDH15	1
PDE1C	1
PKD1L1	1
PLOD3	1
RAD51	2
SCAF4	1
SEL1L	1
SEPTIN14	18
TNRC18	1
TNS3	1
TUT7	1
VOPP1	1
VPS41	1
VSTM2A	1
VWC2	2
YIF1B	1
ZNF713	1
ZBP1	1

2.6 Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_pos_1	protein_pos_2	protein_pos_3	main_ref	chr1_pos	chr2_pos	disrupted_domains	description	annotation	effect_on_fusion
EVT-P-0031404-T01-IM6-2	P-0031404-T01-IM6		BLTP3B-EGFR	BLTP3B	unknown	five_prime	55270272	27	1075	False	retained	1.0	False	Protein tyrosine kinase	UHRF1BP1L	EGFR	NA		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	gene_breakpoint	exon_breakpoint	protein_position	protein_position	main_retained	donor_insertion	unintended_deletion	float_in_domain	disrupted_domains	sequence_annotation	transcript_effect_on_cancer
EVT-P-0050679-T02-IM6-3	P-0050679-T02-IM6		CADPS-EGFR	CADPS	out-of-frame	three_prime	55241099	17	687	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			CADPS (NM_003716) - EGFR (NM_005228) fusion: t(3;7)(p14.2;p11.2)(chr3:g.62537410::chr7:g.55241099::c)	NA
EVT-P-0038683-T01-IM6-4	P-0038683-T01-IM6		CSF2RA-EGFR	CSF2RA	unknown	three_prime	55224220	9	336	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			{CADPS:EGFR} CSF2RA NA (NM_001161530) - EGFR (NM_005228) rearrangement: t(7;X)(p11.2;p22.33)(chr7:g.55224220::chrX:c.435+4059::c) Fusion: mid-exon {CSF2RA:EGFR} DDC NA (NM_000790) - EGFR (NM_005228) Rearrangement: c.435+4059::DDC_c.889+359:EGFRinv Protein Fusion: out of frame {DDC:EGFR}	NA
EVT-P-0016795-T01-IM6-5	P-0016795-T01-IM6		DDC-EGFR	DDC	out-of-frame	three_prime	55222204	7	297	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			DDC NA (NM_000790) - EGFR (NM_005228) Rearrangement: c.435+4059::DDC_c.889+359:EGFRinv Protein Fusion: out of frame {DDC:EGFR}	NA
EVT-P-0018389-T02-IM6-6	P-0018389-T02-IM6		EEA1-EGFR	EEA1	in-frame	three_prime	55241575	18	688	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EEA1 NA (NM_003566) - EGFR (NM_005228) rearrangement: t(7;12)(p11.2;q22)(chr7:g.55241575::chr12:g.100541267::c) Protein Fusion: in frame {EEA1:EGFR}	NA
EVT-P-0064660-T01-IM7-9	P-0064660-T01-IM7		EGFR-CDH7	CDH7	out-of-frame	five_prime	55254206	20	823	True	disrupted	0.438735177	0.5561267	Protein tyrosine and serine/threonine kinase	Protein tyrosine and serine/threonine kinase		(NM_005228) - CDH7 (NM_033646) rearrangement: t(7;18)(p11.2;q22.1)(chr7:g.55254206::chr18:g.55254206::c) Protein Fusion: out of frame {EGFR:CDH7}	NA
EVT-P-0034892-T01-IM6-497	P-0034892-T01-IM6		EGFR-ELAPOR2	ELAPOR2	unknown	three_prime	55268103	24	981	False	lost	0.0	False		Protein tyrosine and serine/threonine kinase		EGFR NA (NM_005228) rearrangement: c.2943:EGFR_chr7:g.86587163del Antisense Fusion EGFR NA (NM_005228) - LANC2 (NM_018697) rearrangement: c.2625+137::EGFR_c.1008+3812:LANCL2 Protein Fusion: in frame {EGFR:LANCL2}	NA
EVT-P-0059222-T03-IM7-500	P-0059222-T03-IM7		EGFR-LANCL2	LANCL2	in-frame	five_prime	55259704	21	875	True	disrupted	0.644268774	0.7036574		Protein tyrosine and serine/threonine kinase		EGFR NA (NM_005228) - LANC2 (NM_018697) rearrangement: c.2625+137::EGFR_c.1008+3812:LANCL2 Protein Fusion: in frame {EGFR:LANCL2}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_is_lost	breakpoint_4	breakpoint_5	main_ref	is_in_frame	is_retained	is_disrupted	description	annotation	effect_on_fra
EVT-P-0037526-T01-IM6-509	P-0037526-T01-IM6		EGFR-SEPTIN14	SEPTIN14	unknown	five_prime	55268916	25	994	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase				EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.2982:EGFR_c.1119+151:SEPT14inv Protein Fusion: mid-exon {EGFR:SEPT14} EGFR NA (NM_005228) - SEPT14 (NM_207366) fusion: c.3271+489:EGFR_c.818-3507:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14} EGFR NA (NM_005228) - SEPT14 (NM_207366) fusion : c.2947-110:EGFR_c.1120-1204:SEPT14 Protein Fusion: in frame {EGFR:SEPT14} EGFR NA (NM_005228) - SEPT14 (NM_207366) fusion : c.2946+306:EGFR_c.1119+2531:SEPT14 Protein Fusion: in frame {EGFR:SEPT14} EGFR NA (NM_005228) - SEPT14 (NM_207366) rearrangement : c.3162+15:EGFR_c.1119+2833:SEPT14 Protein fusion: in frame (EGFR-SEPT14) EGFR NA (NM_005228) - SEPT14 (NM_207366) fusion: c.2947-22:EGFR_c.1120-807:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14} SEPT14 NA (NM_207366) - EGFR (NM_005228) rearrangement: c.1119+1369:SEPT14_c.3163-360:EGFRinv Protein Fusion: in frame {SEPT14:EGFR}	
EVT-P-0039255-T02-IM6-510	P-0039255-T02-IM6		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55270807	27	1091	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.3271+489:EGFR_c.818-3507:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14} EGFR NA (NM_005228) - SEPT14 (NM_207366) fusion : c.2947-110:EGFR_c.1120-1204:SEPT14 Protein Fusion: in frame {EGFR:SEPT14} EGFR NA (NM_005228) - SEPT14 (NM_207366) fusion : c.2946+306:EGFR_c.1119+2531:SEPT14 Protein Fusion: in frame {EGFR:SEPT14} EGFR NA (NM_005228) - SEPT14 (NM_207366) rearrangement : c.3162+15:EGFR_c.1119+2833:SEPT14 Protein fusion: in frame (EGFR-SEPT14) EGFR NA (NM_005228) - SEPT14 (NM_207366) fusion: c.2947-22:EGFR_c.1120-807:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14} SEPT14 NA (NM_207366) - EGFR (NM_005228) rearrangement: c.1119+1369:SEPT14_c.3163-360:EGFRinv Protein Fusion: in frame {SEPT14:EGFR}	
EVT-P-0032604-T01-IM6-512	P-0032604-T01-IM6		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268771	25	983	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				EGFR (NM_005228) - SEPT14 (NM_207366) fusion : c.2947-110:EGFR_c.1120-1204:SEPT14 Protein Fusion: in frame {EGFR:SEPT14} EGFR NA (NM_005228) - SEPT14 (NM_207366) fusion : c.2946+306:EGFR_c.1119+2531:SEPT14 Protein Fusion: in frame {EGFR:SEPT14} EGFR NA (NM_005228) - SEPT14 (NM_207366) rearrangement : c.3162+15:EGFR_c.1119+2833:SEPT14 Protein fusion: in frame (EGFR-SEPT14) EGFR NA (NM_005228) - SEPT14 (NM_207366) fusion: c.2947-22:EGFR_c.1120-807:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14} SEPT14 NA (NM_207366) - EGFR (NM_005228) rearrangement: c.1119+1369:SEPT14_c.3163-360:EGFRinv Protein Fusion: in frame {SEPT14:EGFR}	
EVT-P-0031753-T01-IM6-513	P-0031753-T01-IM6		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268412	24	982	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				EGFR (NM_005228) - SEPT14 (NM_207366) fusion (EGFR exons 1-24 fused to SEPT14 exon 10): c.2946+306:EGFR_c.1119+2531:SEPT14 Protein Fusion: in frame {EGFR:SEPT14} EGFR NA (NM_005228) - SEPT14 (NM_207366) rearrangement : c.3162+15:EGFR_c.1119+2833:SEPT14 Protein fusion: in frame (EGFR-SEPT14) EGFR NA (NM_005228) - SEPT14 (NM_207366) fusion: c.2947-22:EGFR_c.1120-807:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14} SEPT14 NA (NM_207366) - EGFR (NM_005228) rearrangement: c.1119+1369:SEPT14_c.3163-360:EGFRinv Protein Fusion: in frame {SEPT14:EGFR}	
EVT-P-0005701-T01-IM5-514	P-0005701-T01-IM5		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55269490	26	1054	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				EGFR (NM_005228) - SEPT14 (NM_207366) fusion (EGFR exons 1-24 fused to SEPT14 exon 10): c.2946+306:EGFR_c.1119+2531:SEPT14 Protein Fusion: in frame {EGFR:SEPT14} EGFR NA (NM_005228) - SEPT14 (NM_207366) rearrangement : c.3162+15:EGFR_c.1119+2833:SEPT14 Protein fusion: in frame (EGFR-SEPT14) EGFR NA (NM_005228) - SEPT14 (NM_207366) fusion: c.2947-22:EGFR_c.1120-807:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14} SEPT14 NA (NM_207366) - EGFR (NM_005228) rearrangement: c.1119+1369:SEPT14_c.3163-360:EGFRinv Protein Fusion: in frame {SEPT14:EGFR}	
EVT-P-0043525-T01-IM6-515	P-0043525-T01-IM6		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268859	25	983	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.2947-22:EGFR_c.1120-807:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14} SEPT14 NA (NM_207366) - EGFR (NM_005228) rearrangement: c.1119+1369:SEPT14_c.3163-360:EGFRinv Protein Fusion: in frame {SEPT14:EGFR}	
EVT-P-0029159-T01-IM6-516	P-0029159-T01-IM6		EGFR-SEPTIN14	SEPTIN14	in-frame	three_prime	55269850	27	1055	True	lost	0.0	False	Protein tyrosine and serine/threonine kinase	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.2947-22:EGFR_c.1120-807:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14} SEPT14 NA (NM_207366) - EGFR (NM_005228) rearrangement: c.1119+1369:SEPT14_c.3163-360:EGFRinv Protein Fusion: in frame {SEPT14:EGFR}	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	clonally_deleted	inframe	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0028241-T01-IM6-517	P-0028241-T01-IM6		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268525	25	983	True	retained	1.0	False		Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - SEPT14 (NM_207366) fusion (EGFR exons 1-24 fused to SEPT14 exon 10): c.2947-356:EGFR_c.1119+4312:SEPT14inv	NA	
EVT-P-0025191-T02-IM6-518	P-0025191-T02-IM6		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55269542	26	1054	True	retained	1.0	False		Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.3162+67:EGFR_c.1120-3396:SEPT14inv	NA	
EVT-P-0061245-T01-IM7-519	P-0061245-T01-IM7		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268815	25	983	True	retained	1.0	False		Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - SEPT14 (NM_207366) rearrangement: c.2947-66:EGFR_c.1119+2973:SEPT14inv	NA	
EVT-P-0014220-T02-IM6-520	P-0014220-T02-IM6		EGFR-SEPTIN14	SEPTIN14	unknown	five_prime	55268942	25	1003	False	retained	1.0	False		Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - SEPT14 (NM_207366) fusion (EGFR exons 1-25 fused with SEPT14 exon 10): c.3008:EGFR_c.1120-4102:SEPT14inv	NA	
EVT-P-0023043-T01-IM6-521	P-0023043-T01-IM6		EGFR-SEPTIN14	SEPTIN14	in-frame	five_prime	55268179	24	982	True	retained	1.0	False		Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - SEPT14 (NM_207366) Fusion (EGFR exons 1-24 fused with SEPT14 exon 10) : c.2946+73:EGFR_c.1120-1180:SEPT14	NA	

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3 Figures

EGFR domain-retention track

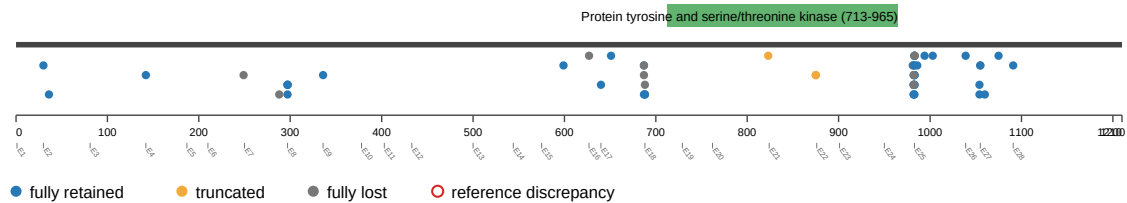
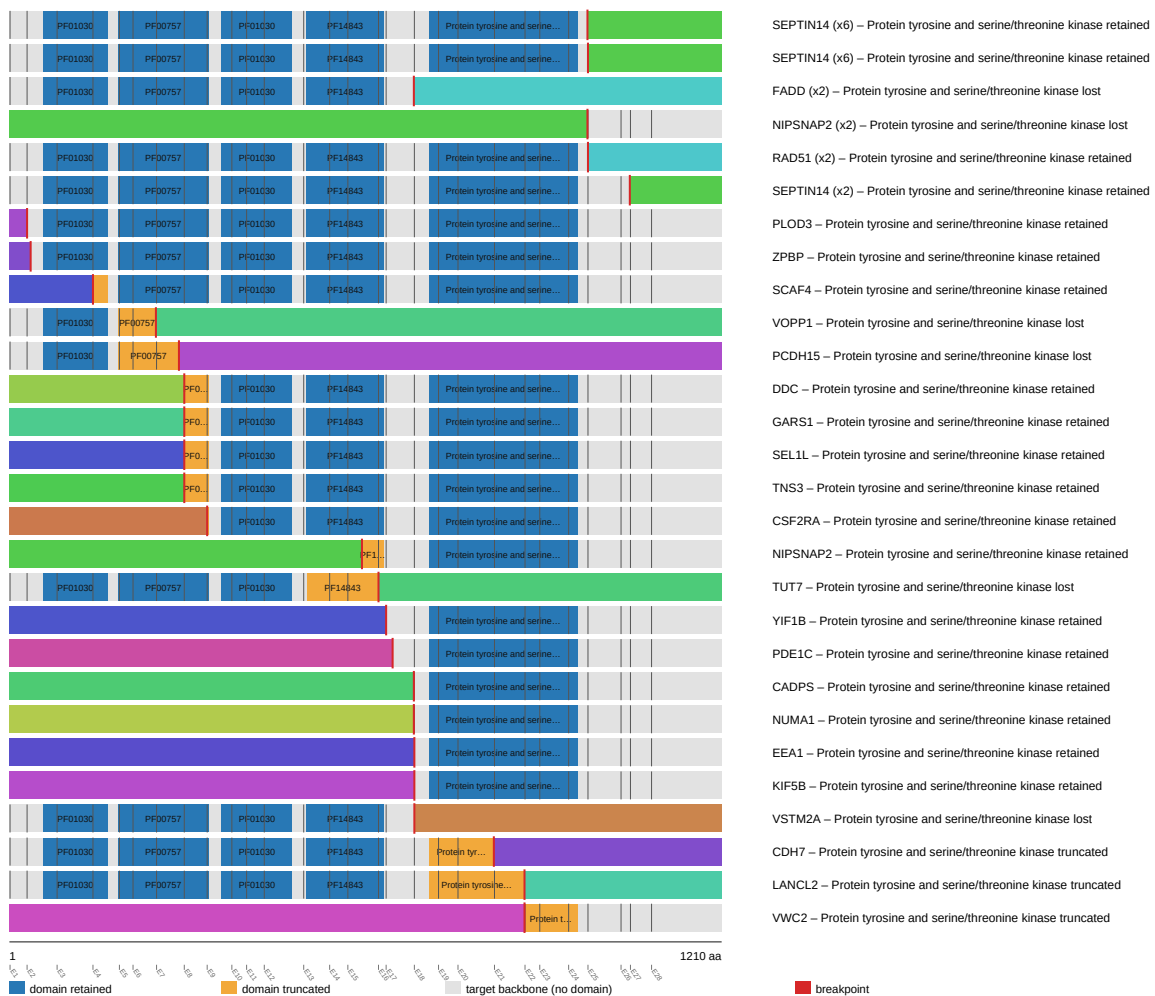


Figure 1: domain retention outliers

EGFR fusion-transcript schematic

partner block → breakpoint → retained EGFR portion (domain-colored, exon ticks)



Showing the top 28 of 41 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

Figure 2: fusion schematic

EGFR intragenic-deletion schematic

retained N-terminal block → deleted span (plain connector) → resumed C-terminal block

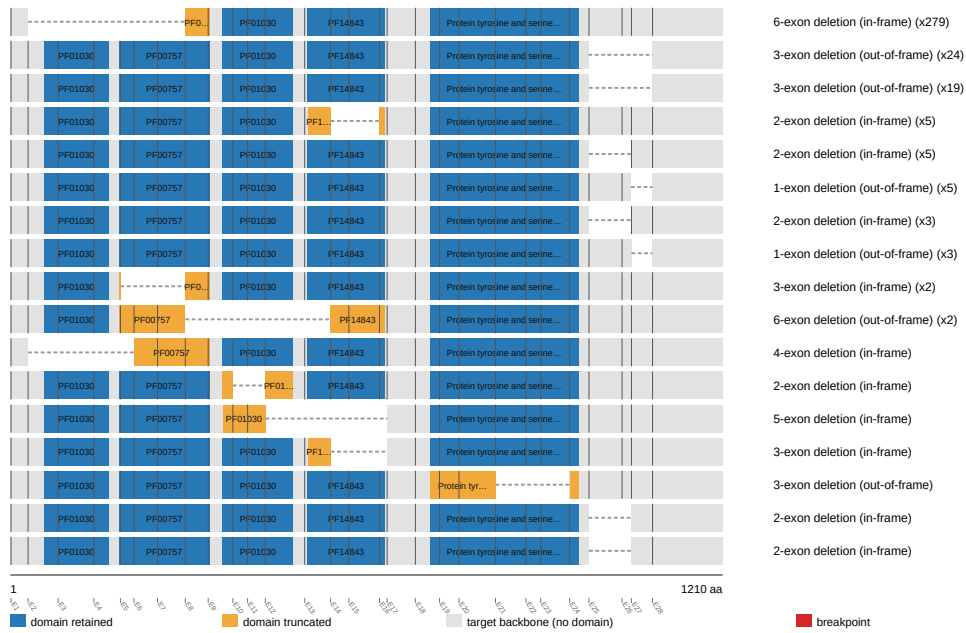


Figure 3: intragenic deletion schematic

Reference vs msk_impact_50k_2026

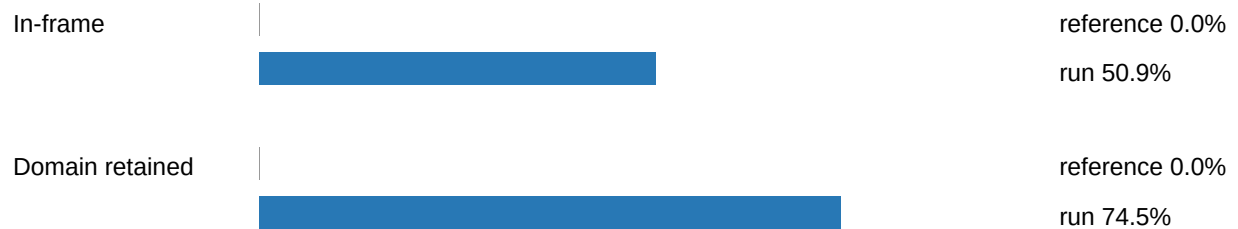


Figure 4: reference comparison